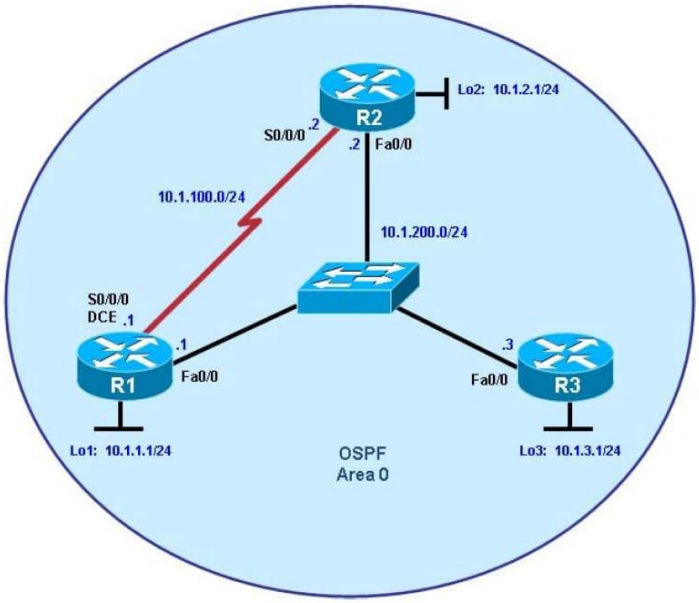
Practical 1: Single Area OSPF Configuration (Area 0)

Devices Used:

- 3 Routers (Cisco 2811)
- 1 Switch (1841)+add HWIC 2T
- Serial and FastEthernet connections



Practical 1 – Single Area OSPF

R1

Copy-paste this block in R1

```
enable
configure terminal
interface Loopback1
description Engineering Department
ip address 10.1.1.1 255.255.255.0
exit
interface FastEthernet0/0
ip address 10.1.200.1 255.255.255.0
no shutdown
exit
interface Serial0/0/0
```

```
ip address 10.1.100.1 255.255.255.0
clock rate 64000
bandwidth 64
no shutdown
exit
router ospf 1
network 10.1.100.0 0.0.0.255 area 0
network 10.1.200.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.255 area 0
exit
interface Loopback1
ip ospf network point-to-point
exit
interface FastEthernet0/0
ip ospf cost 50
ip ospf priority 10
end
```

R2

□ Copy-paste this block in R2

```
enable
configure terminal
interface Loopback2
description Marketing Department
ip address 10.1.2.1 255.255.255.0
exit
interface FastEthernet0/0
ip address 10.1.200.2 255.255.255.0
no shutdown
exit
interface Serial0/0/0
ip address 10.1.100.2 255.255.255.0
bandwidth 64
no shutdown
exit
router ospf 1
network 10.1.100.0 0.0.0.255 area 0
network 10.1.200.0 0.0.0.255 area 0
network 10.1.2.0 0.0.0.255 area 0
exit
interface Loopback2
ip ospf network point-to-point
exit
interface FastEthernet0/0
ip ospf cost 50
ip ospf priority 5
end
```

R3

□ Copy-paste this block in R3

```
enable
configure terminal
interface Loopback3
description Accounting Department
ip address 10.1.3.1 255.255.255.0
exit
interface FastEthernet0/0
ip address 10.1.200.3 255.255.255.0
no shutdown
exit
router ospf 1
network 10.1.200.0 0.0.0.255 area 0
network 10.1.3.0 0.0.0.255 area 0
exit
interface Loopback3
ip ospf network point-to-point
exit
interface FastEthernet0/0
ip ospf cost 50
end
```

Verification Commands (run when required)

```
show ip protocols
show ip ospf
show ip ospf neighbor
show ip ospf interface brief
show ip ospf database
show ip route
show ip ospf neighbor detail
```